



ANDHRA LOYOLA
INSTITUTE OF ENGINEERING & TECHNOLOGY
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A Competition
On

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Organized By

DEPARTMENT OF

ELECTRICAL & ELECTRONICS ENGINEERING

(In Association with IEI STUDENTS' CHAPTER)

**DUAL-AXIS SOLAR TRACKING AND MONITORING
SYSTEM FOR ENHANCED ENERGY EFFICIENCY**

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Objective

- ➤ Design a dual-axis solar tracking system that automatically follows the sun
- ➤ Increase solar panel efficiency compared with fixed panels
- ➤ Use ESP8266 NodeMCU to control sensors and servo motors
- ➤ Implement real-time monitoring through a web interface
- ➤ Maintain maximum sunlight exposure

Literature Survey

- ➤ Fixed solar panels capture limited sunlight due to sun movement
- ➤ Single-axis trackers improve performance but still have limitations
- ➤ Dual-axis systems increase energy harvesting efficiency
- ➤ Microcontrollers enable automatic tracking
- ➤ IoT systems allow remote monitoring and control

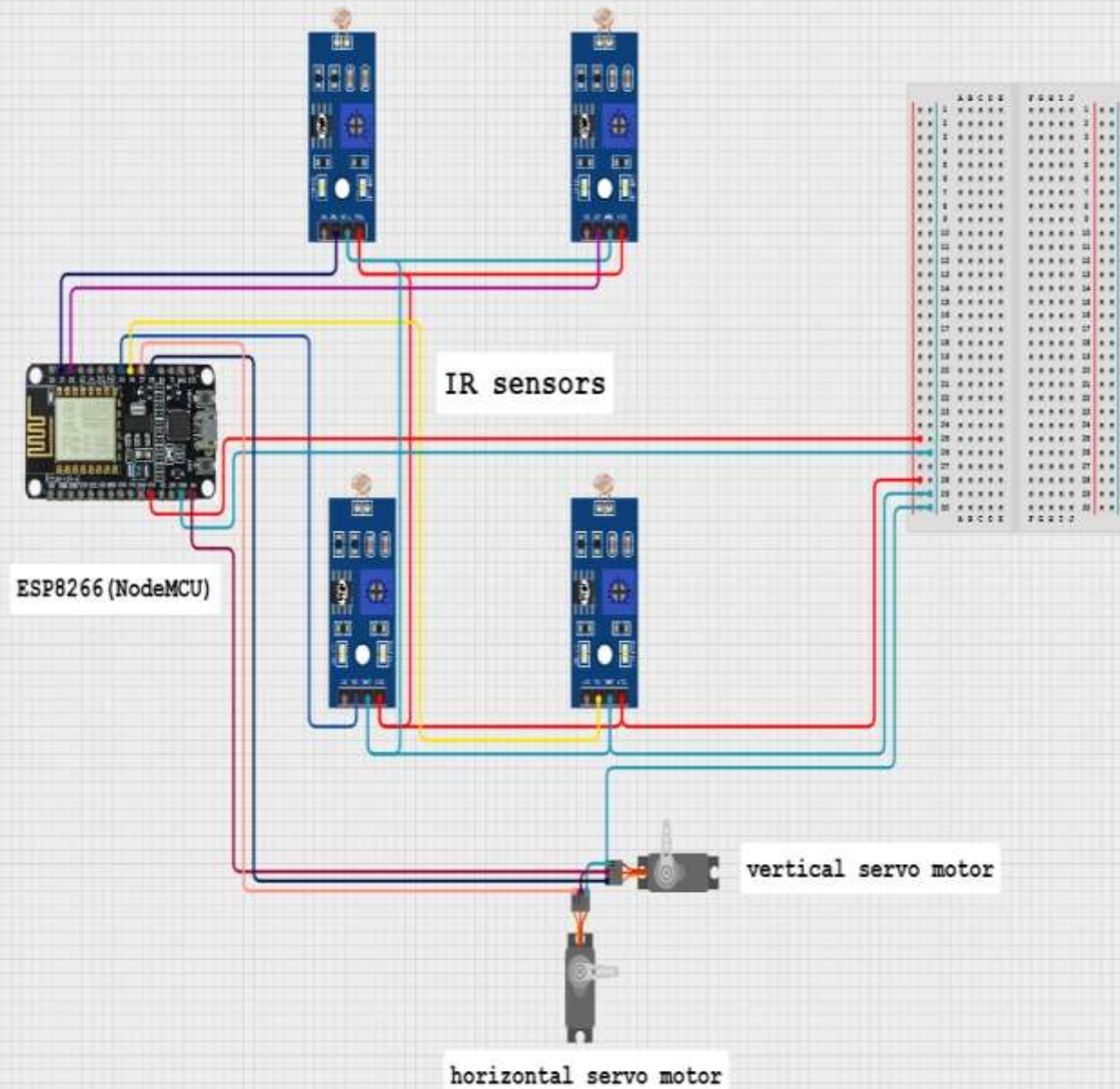
Introduction

- ➤ Solar energy is a major renewable energy source
- ➤ Efficiency depends on the angle between sun and panel
- ➤ Fixed panels cannot adjust to sun movement
- ➤ Dual-axis tracking adjusts horizontal and vertical directions
- ➤ Improves power generation compared to static systems

Methodology

- ➤ LDR sensors detect sunlight intensity
- ➤ ESP8266 reads sensor values
- ➤ Compare top-bottom and left-right sensor values
- ➤ Servo motors rotate panel based on light difference
- ➤ Continuous adjustment ensures maximum sunlight
- ➤ Data displayed on web dashboard

DUAL AXIS SOLAR MONITORING SYSTEM



Results

- ➤ Solar panel continuously aligns with sun
- ➤ Higher energy capture compared to fixed panels
- ➤ Real-time monitoring of light intensity and servo angles
- ➤ Fast response to sunlight direction changes



Conclusion & Future Scope

- Conclusion:
 - ► System successfully tracks sun in two axes
 - ► Improves solar energy efficiency
 - ► IoT monitoring increases accessibility
- Future Scope:
 - ► Battery monitoring integration
 - ► AI-based predictive tracking
 - ► Application in large solar farms

References

- ➤ Solar Energy Engineering – Duffie & Beckman
- ➤ IEEE papers on Solar Tracking Systems
- ➤ ESP8266 NodeMCU Documentation
- ➤ Renewable Energy Journals

Thank You

- Questions?